

BANCO PRELIMINAR DE CASOS

Toma de decisiones asistida con IA

El estudio seguirá un diseño experimental comparativo y exploratorio, donde se evaluará el comportamiento de distintos modelos de IA generativa frente a un conjunto estandarizado de problemas de decisión multicriterio.

1. Casos

Se desarrollarán, aproximadamente, ocho casos de decisión distribuidos en cuatro niveles de complejidad (dos casos por nivel), buscando representar distintos contextos de toma de decisiones y permitir la comparación sistemática entre modelos de IA.

Nivel Características

Nivel 1 Información completa y criterios objetivos.

Nivel 2 Múltiples preferencias y compensaciones (trade-offs).

Nivel 3 Riesgo con probabilidades conocidas.

Nivel 4 Alta incertidumbre e información incompleta.

2. Prompts

Cada caso tendrá dos prompts base estandarizados. El primer tipo será ambiguo, sólo con los datos primordiales; el segundo tipo serán más específicos con los datos, preferencias y detalles. Después, si el periodo de tiempo alcanza, se generarán variaciones para estudiar el efecto de *framing* (el cómo el contexto, las palabras moldean la manera en que se percibe, o se entiende). Como ejemplo de las variaciones sería cambiar el orden de la información, el orden de las alternativas, modificar ligeramente la redacción, presentar primero los beneficios o riesgos.

En cuanto a la metodología de los prompts, ambos, tanto el tipo uno y dos, se clasifican como *zero-shot prompting*, es decir, se proveen cero ejemplos a la Inteligencia Artificial; se les da directamente el escenario del problema y se pide la recomendación. Cabe recalcar que, pese a que se utiliza el mismo método para los dos, el tipo uno es un *basic zero-shot*, en cambio al tipo dos que es un *contextualised zero-shot*.

Prompt 1 (Ambiguous).

Prompt 2 (Specific).

Al omitir ejemplos previos de resolución (*zero-shot*), se obliga al modelo de IA a depender exclusivamente de sus capacidades nativas de cálculo, lógica y análisis de decisión bajo incertidumbre.

Level 1. Problemas con información completa y criterios objetivos.

Problem 1. This case was taken from a paper “The best vacuum robot cleaners with the help of the COMET and the TOPSIS methods”

Table 1. Chosen criteria C_1 - C_8 and their respective characteristic values for linguistic variables low, high

C_i	Name	Type	Unit
C_1	Engine power	Profit	Watt [W]
C_2	Noise level	Cost	Decibel [dB]
C_3	Bag capacity	Profit	Litre [L]
C_4	Water tank capacity	Profit	Litre [L]
C_5	Charging time	Cost	Hour [h]
C_6	Working time	Profit	Minutes [min]
C_7	Battery capacity	Profit	Mega ampere hour [mAh]
C_8	Price	Cost	Polish zloty [PLN]

Table 2. Performance alternatives set

Alternatives	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
Ecovacs Deebot U2	26	66.0	0.40	0.30	4.5	110	2600	899.0
Ecovacs Deebot OZMO T8+	40	67.0	0.42	0.24	6.5	175	5200	3199.0
Ecovacs Deebot OZMO 950	40	67.0	0.43	0.24	5.0	200	5200	1899.0
Ecovacs Deebot OZMO T8 Aivi	40	67.0	0.42	0.24	6.5	175	5200	2899.0
Roborock S5 MAX	58	67.0	0.46	0.30	5.0	180	5200	699.0
Roborock E4	58	66.0	0.64	0.18	6.0	200	5200	959.0
Karcher RC 3	24	71.0	0.35	0.00	4.0	120	2600	2229.0
Kiano Elegance Robot	30	65.0	0.35	0.35	5.0	120	4400	1199.0
Black&Decker RVA425B-QW	36	65.0	1.00	0.00	4.0	90	2500	1399.0
Roborock E4	58	69.0	0.64	0.18	6.0	150	5200	949.0
Xiaomi MI Robot Vacuum-Mop Pro	33	78.0	0.55	0.20	3.0	90	3200	1288.0
Ariete Evolution 2712	25	67.6	0.50	0.00	3.0	90	2600	999.0

Table 3. Normalized decision matrix

A_i	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
A_1	0.05882	0.92308	0.07692	0.85714	0.57143	0.18182	0.03704	0.9200
A_2	0.47059	0.84615	0.10769	0.68571	0.00000	0.77273	1.00000	0.0000
A_3	0.47059	0.84615	0.12308	0.68571	0.42857	1.00000	1.00000	0.5200
A_4	0.47059	0.84615	0.10769	0.68571	0.00000	0.77273	1.00000	0.1200
A_5	1.00000	0.84615	0.16923	0.85714	0.42857	0.81818	1.00000	1.0000
A_6	1.00000	0.92308	0.44615	0.51429	0.14286	1.00000	1.00000	0.8960
A_7	0.00000	0.53846	0.00000	0.00000	0.71429	0.27273	0.03704	0.3880
A_8	0.17647	1.00000	0.00000	1.00000	0.42857	0.27273	0.70370	0.8000
A_9	0.35294	1.00000	1.00000	0.00000	0.71429	0.00000	0.00000	0.7200
A_{10}	1.00000	0.69231	0.44615	0.51429	0.14286	0.54545	1.00000	0.9000
A_{11}	0.26471	0.00000	0.30769	0.57143	1.00000	0.00000	0.25926	0.7644
A_{12}	0.02941	0.80000	0.23077	0.00000	1.00000	0.00000	0.03704	0.8800

Table 1. Chosen crit

Table 4. The 12 robot vacuum cleaners result obtained from the model sequentially

Alternatives	$COMET_{pref}$	$TOPSIS_{pref}$	$COMET_{rank}$	$TOPSIS_{rank}$
1	0.4533	0.4705	9	9
2	0.4854	0.4907	7	7
3	0.6343	0.5996	4	4
4	0.5004	0.5002	6	6
5	0.7649	0.6883	1	1
6	0.7403	0.6659	2	2
7	0.2438	0.3092	12	12
8	0.5477	0.5315	5	5
9	0.4734	0.4842	8	8
10	0.6551	0.6155	3	3
11	0.3959	0.4286	10	10
12	0.3722	0.4252	11	11

The best alternative is A5.

Prompts.

Prompt 1. Oye, quiero comprarme una aspiradora robot para mi casa. Tengo varias opciones: Ecovacs Deebot U2, Ecovacs Deebot OZMO T8, Ecovacs Deebot 950, Ecovacs Deebot OZMO T8 Aivi,, Robot Rock S5 MAX, Robot Rock E4, Karcher RC 3, Kiano Elegance Robot, Black & Decker RVA425B-QW, Robot Rock E4, Xiaomi MI Robot Vacuum-Mop Pro, Ariete Evolution 2712, pero no sé por cuál decidirme. ¿Me ayudas a elegir la mejor?

Prompt 2. Hola. Estoy comparando varios modelos de robots aspiradores porque no sé cuál me conviene más. Tengo una lista con opciones: Ecovacs Deebot U2, Ecovacs Deebot OZMO T8, Ecovacs Deebot 950, Ecovacs Deebot OZMO T8 Aivi,, Robot Rock S5 MAX, Robot Rock E4, Karcher RC 3, Kiano Elegance Robot, Black&Decker RVA425B-QW, Robot Rock E4, Xiaomi MI Robot Vacuum-Mop Pro, Ariete Evolution 2712. Para decidirme quiero tomar en cuenta cosas fijas como la potencia del motor, qué tanto ruido hacen, la capacidad del tanque de agua y de la bolsa, cuánto tarda en cargar, cuánto le dura la batería en uso y, el precio. ¿Cuál opción me recomiendas?

Nota. En ambos prompts se adjuntará la siguiente tabla de información para que sea más amigable con la IA.

<i>Modelo</i>	<i>Potencia (W)</i>	<i>Ruido (dB)</i>	<i>Bolsa (L)</i>	<i>Tanque de agua (L)</i>	<i>Carga (h)</i>	<i>Autonomía (min)</i>	<i>Batería (mAh)</i>	<i>Precio (PLN)</i>
Ecovacs Deebot U2	26	66.0	0.40	0.30	4.5	110	2600	899.0
Ecovacs Deebot OZMO T8+	40	67.0	0.42	0.24	6.5	175	5200	3199.0
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Problem 2. *Choosing a car, a case study proposed and designed by us.*

Suppose that a person wants to buy a car and has the exact money, the prices are fixed, he does not care about the color, the brand or the status.

Alternatives.

A1 (Renault Kwid), A2 (Suzuki Swift Boostergreen), and A3 (Chevrolet Aveo Sedán).

Improved safety and functionality (airbags, reversing camera, boot size) combined with the best performance (fuel efficiency, power and tyre size).

NOTA: pondré que el objetivo será diferente al de reducción de costo para observar en el futuro si la AI alucina o sesga de forma irracional recomendando el que más barato.

Criteria.

<i>Ci</i>	<i>Name</i>	<i>Unit</i>
C1	Price	MXN
C2	Fuel efficiency (Combined)	Kilometers per liter (km/L)
C3	Engine power	Horsepower (HP)
C4	Airbags quantity	Units
C5	Rear view camera	Binary indicador (0/1)
C6	Boot capacity	Litre
C7	Wheel size	Inches

Data.

Alternatives	C1	C2	C3	C4	C5	C6	C7
A1	251,500	22.1	66	4	0	290	14''
A2	339,990	24.7	82	6	1	242	16''
A3	261,500	20.4	98	6	1	475	15''

Possible reference solution using the SMART methodology for MCDA.

1. Assignment of relative weights.

C_j	Importance (0-100)	W_j
C4	100	0.220
C5	90	0.198
C6	85	0.187
C2	65	0.143
C3	55	0.121
C7	45	0.999
C1	15	0.033
total	455	1.000

2. Attribute normalization.

	Car A	Car B	Car C
C1	100.0	0.0	88.7
C2	39.5	100.0	0.0
C3	0.0	50.0	100.0
C4	0.0	100.0	100.0
C5	0.0	100.0	100.0
C6	20.6	0.0	100.0
C7	0.0	100.0	50.0

3. Global value.

Using

$$V(A_i) = \sum_{j=1}^n w_j \cdot V_j(x_{ij})$$

DECISION MATRIX

Alternative	SMART method	Rank
A1	12.80	3
A2	72.05	2
A3	80.48	1

The optimal decision would be A3, and the secondary alternative would be A2.

Prompts.

Prompt 1. Hola. Estoy por comprar un auto y mi objetivo principal es la seguridad y funcionalidad con el mejor desempeño. Estoy entre el Car A (Renault Kwid a \$251,500), Car B (Suzuki Swift Boostergreen a \$339,990) y Car C (Chevrolet Aveo Sedán a \$261,500). Te voy a dar unos datos de cada uno, ¿cuál me recomiendas?

Prompt 2. Hola, necesito decidir qué coche comprar entre tres opciones específicas: el Renault Kwid, el Suzuki Swift Boostergreen y el Chevrolet Aveo Sedán. Ya tengo el dinero exacto y los precios son fijos, así que el costo no es mi prioridad número uno. Lo que realmente me importa es que tenga la mejor seguridad, las mayores funciones para su uso (si tiene cámara de reversa y el tamaño de la cajuela) combinado con un buen rendimiento de gasolina, potencia y tamaño de las llantas. Te paso los datos de rendimiento, caballos de fuerza y espacio de cada uno para que me ayudes a decidir.

Nivel 2. Problemas con preferencias y compensaciones entre objetivos.

Problem 1. “The summer job” an example from the book p. 128 *Making Hard Decisions an Introduction to Decision Analysis*.

Assume that Sam Chu was in a quandary. With two job offers in hand, the choice he should make was far from obvious.

The first alternative was a job as an assistant at a local small business; the job would pay minimum wage (\$5.25 per hour), require 25 to 35 hours per week, and the hours would be primarily during the week, leaving the weekends free. The job would last for three months, but the exact amount of work, and hence the amount Sam could earn, was uncertain. On the other hand, the free weekends could be spent with friends.

The second alternative was to work as a member of a trail-maintenance crew for a conservation organization. This job would require 10 weeks of hard work, 40 hours per week at \$6.50 per hour, in a national forest in a neighboring state. The job would involve extensive camping and backpacking. Members of the maintenance crew would come from a large geographic area and spend the entire 10 weeks together, including weekends. Although Sam had no doubt about the earnings this job would provide, the real uncertainty was what the staff and other members of the crew would be like. Would new friendships develop? The nature of the crew and the leaders could make for 10 weeks of a wonderful time, 10 weeks of misery, or anything in between.

From the description, it appears that Sam has two objectives in this context: earning money and having fun this summer. Both are reasonable, and the two jobs clearly differ in these two dimensions; they offer different possibilities for the amount of money earned and the quality of summer fun.

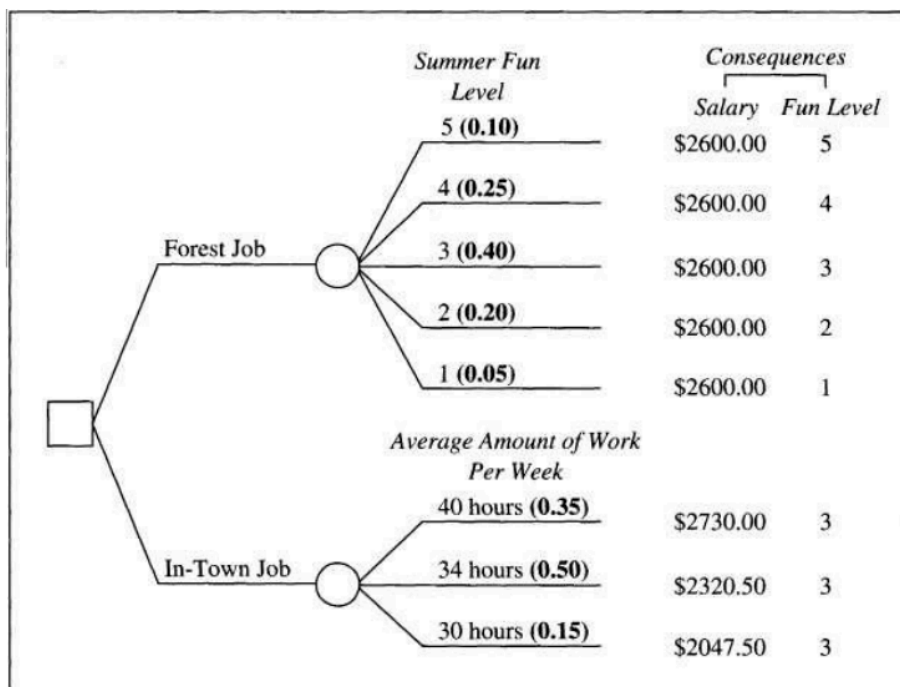
The amount of money to be earned has a natural scale (dollars), and like most of us Sam prefers more money to less. The objective of having fun has no natural scale, though. Thus, the first step is to create such a scale. After considering the possibilities, Sam has created the scale in Table 4.5 to represent different levels of summer fun in the context of choosing a summer job. Although living in town and living in a forest camp pose two very different scenarios, the scale has been constructed in such a way that it can be applied to either job (as well as to any other prospect that might arise).

The levels are numbered so that the higher numbers are more preferred.

Table 4.5

A constructed scale for summer fun.

- 5 (Best) A large, congenial group. Many new friendships made. Work is enjoyable, and time passes quickly.
- 4 A small but congenial group of friends. The work is interesting, and time off work is spent with a few friends in enjoyable pursuits.
- 3 No new friends are made. Leisure hours are spent with a few friends doing typical activities. Pay is viewed as fair for the work done.
- 2 Work is difficult. Coworkers complain about the low pay and poor conditions. On some weekends it is possible to spend time with a few friends, but other weekends are boring.
- 1 (Worst) Work is extremely difficult, and working conditions are poor. Time off work is generally boring because outside activities are limited or no friends are available.



The tree reflects Sam's belief that summer fun with the in-town job will amount to Level 3 in the constructed scale, but there is considerable uncertainty about how much fun the forest job will be. This uncertainty has been translated into probabilities based on Sam's uncertainty; how to make such judgments is the topic of Chapter 8. Likewise, the decision tree reflects uncertainty about the amount of work available at the in-town job.

Prompts.

Prompt 1. Tengo dos ofertas de trabajo para este verano: uno en un negocio local de mi ciudad y otro arreglando senderos en un bosque nacional. No sé cuál aceptar, ¿qué me recomiendas?

Prompt 2. Hola, tengo dos ofertas de trabajo de verano y no sé qué hacer porque mis metas son ganar dinero y divertirme. El primer trabajo es en mi ciudad; me pagan el salario mínimo (\$5.25 la hora), trabajo entre 25 y 35 horas por tres meses, y tengo los fines de semana libres para estar con mis amigos, aunque las horas exactas de trabajo semanal son algo inciertas; hay una probabilidad de 35% que trabaje 40 horas, 50% por 34 horas y 15% por 30 horas El segundo trabajo es en un bosque nacional en otro estado manteniendo senderos; son 10 semanas fijas de 40 horas a \$6.50 la hora (ganaría seguro \$2600), pero implica acampar todo el tiempo y convivir con desconocidos las 24 horas del día, lo que me da miedo es de que el ambiente sea increíble o una completa miseria. En el de la ciudad sé que la diversión será un nivel medio, pero en el bosque hay mucha incertidumbre. Evalúe los posibles niveles de diversión, 5-4 (disfrutar el trabajo y hacer nuevos amigos) con 35% de probabilidad, 3 (sólo me pagan por el trabajo hecho, no hice nuevos amigos) 40%, y 2-1 (Las condiciones laborales son horribles y es aburrido) con 25%¿Cuál me recomiendas...?

Problem 2. Choosing a university, a case study proposed and designed by us.

Alternatives.

A1 (Hochschule für Wirtschaft und Recht Berlin),

A2 (University of Potsdam).

C_j	Name	Description
C1	Professional immersion & networking	Opportunities for events, job fairs, and internships. (Scale 1-10, maximize)
C2	Cost of living & accomodation	Estimated monthly budget in euros. (Scale in €, minimize).
C3	Student quality of life	Balance between tranquility, culture, and ease of integration. (Scale 1-10, maximize).

Criteria.

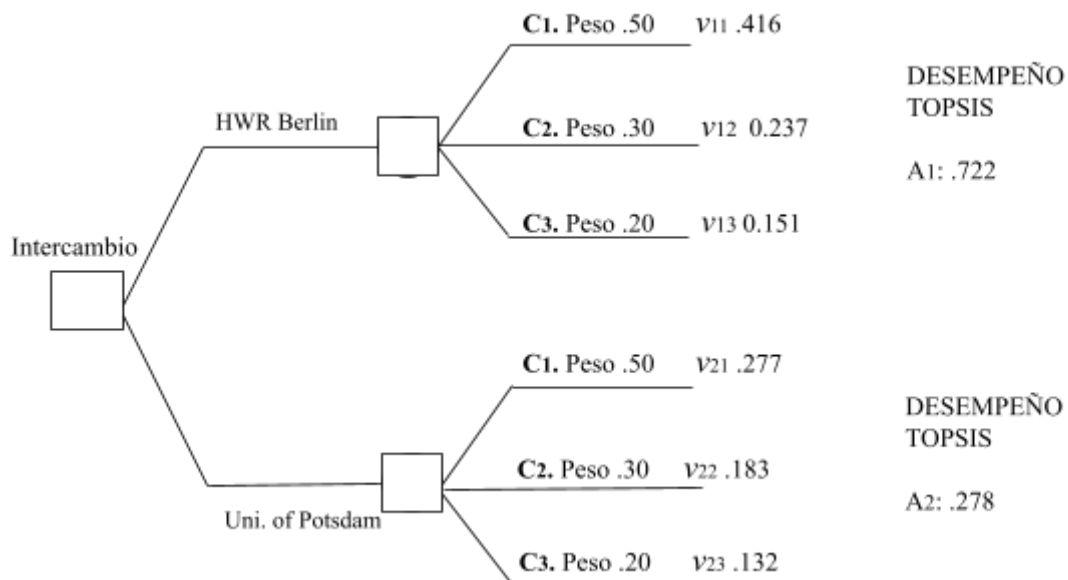
C_i	A1	A2
C1	9/10	6/10
C2	1,100 €	850 €
C3	8/10	7/10

C_i	W_j
C1	0.50
C2	0.30
C3	0.20

The trade-off in this example is that A1 offers a fairly significant advantage in terms of networking, but at an extra cost of €250 per month. So, is it worth paying 29% more of your budget for that professional advantage?

A possible solution using the TOPSIS method

DECISIÓN TREE.



Euclidean normalisation for each criterion.

$$C1 \sqrt{9^2 + 6^2} = 10.817$$

$$C2 \sqrt{1100^2 + 850^2} = 1390.144$$

$$C3 \sqrt{8^2 + 7^2} = 10.63$$

Multiplying each normalised value by its respective weighting factor ($v_{ij} = r_{ij} \cdot w_j$)

A1

$$v_{11} = \left(\frac{9}{10.817}\right) \cdot 0.50 = 0.416$$

$$v_{12} = \left(\frac{1100}{1390.144}\right) \cdot 0.30 = 0.237$$

$$v_{13} = \left(\frac{8}{10.63}\right) \cdot 0.20 = 0.151$$

A2

$$v_{21} = \left(\frac{6}{10.817}\right) \cdot 0.50 = 0.277$$

$$v_{22} = \left(\frac{850}{1390.144}\right) \cdot 0.30 = 0.183$$

$$v_{23} = \left(\frac{7}{10.630}\right) \cdot 0.20 = 0.132$$

The best maximum values are C1 and C3, and the best minimum value is C2.

$$A^+ \{(0.416, 0.277), (0.237, 0.183), (0.151, 0.132)\}$$

$$\{0.416, 0.183, 0.151\}$$

A negative ideal implies the worst values.

$$A^- \{(0.416, 0.277), (0.237, 0.183), (0.151, 0.132)\}$$

$$\{0.277, 0.237, 0.132\}$$

$$S_1^+ = \sqrt{(0.416 - 0.416)^2 + (0.237 - 0.183)^2 + (0.151 - 0.151)^2} = 0.0540$$

$$S_1^- = \sqrt{(0.416 - 0.277)^2 + (0.237 - 0.237)^2 + (0.151 - 0.132)^2} = 0.1403$$

$$S_2^+ = \sqrt{(0.277 - 0.416)^2 + (0.183 - 0.183)^2 + (0.132 - 0.151)^2} = 0.1403$$

$$S_2^- = \sqrt{(0.277 - 0.277)^2 + (0.183 - 0.237)^2 + (0.132 - 0.132)^2} = 0.054$$

$$C_i = \frac{S_i^-}{S_i^+ + S_i^-}$$

formula for determining.

$$A1 \frac{0.1403}{0.054+0.1403} = 0.722$$

$$A2 \frac{0.054}{0.1403+0.054} = 0.278$$

Using the TOPSIS method, alternative A1 is the optimal choice.

Prompts.

Prompt 1. Me voy a ir de intercambio a Alemania y tengo que elegir entre la Hochschule für Wirtschaft und Recht Berlin y la Universidad de Potsdam. ¿Cuál me recomiendas elegir?

Prompt 2. Hola, tengo que decidir a qué universidad irme de intercambio, si a la HWR en Berlín o a la Universidad de Potsdam. Tengo un dilema con mis prioridades. Hice una evaluación de las ventajas profesionales, del costo de vida en euros, y de la calidad de vida. Berlín me da nivel de networking e inmersión profesional un 9/10 frente a un 6/10 de

Potsdam; de calidad de vida estudiantil le doy un 8/10 a la de Berlín y 7/10 a la de Potsdam. El problema es el dinero, vivir en Berlín me cuesta unos 1,100 euros al mes, mientras que en Potsdam gastaría unos 850 euros. O sea, Berlín es un 29% más caro. Quiero que me ayudes a decidir si crees que vale la pena pagar ese extra, ¿cuál de las dos opciones me recomiendas?

Nivel 3. Problemas con riesgo y probabilidades.

Problem 1. “Risk” an example from the book p. 463 *Making Hard Decisions an Introduction to Decision Analysis*.

Imagine that you have the opportunity to play one game or the other, but only one time. Which one would you prefer to play? Your choice also is drawn in decision-tree form in Figure 13.1.

Alternatives.

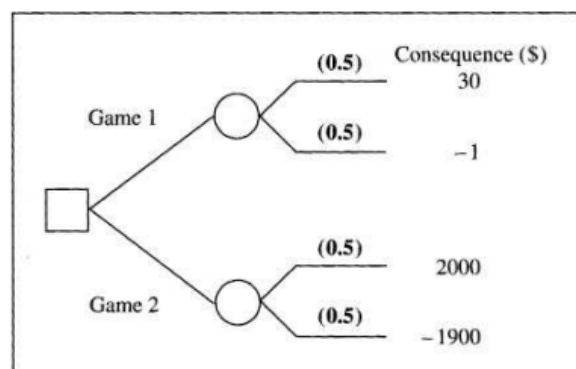
Game 1 Win \$30 with probability 0.5 Lose \$1 with probability 0.5

Game 2 Win \$2000 with probability 0.5 Lose \$1900 with probability 0.5

Game 1 has an expected value of \$14.50. Game 2, on the other hand, has an expected value of \$50.00. If you were to make your choice on the basis of expected value, then you would choose Game 2. Most of us, however, would consider Game 2 to be riskier than Game 1, and it seems reasonable to suspect that most people actually would prefer Game 1. Using expected values to make decisions means that the decision maker is considering only the average or expected payoff. If we take a long-run frequency approach, the expected value is the average amount we would be likely to win over many plays of the game. But this ignores the range of possible values. After all, if we play each game 10 times, the worst we could do in Game 1 is to lose \$10. On the other hand, the worst we could do in Game 2 is lose \$19,000!

DECISION TREE

Figure 13.1
Two lottery games.
Which game would you choose?



Prompts.

Prompt 1. Me ofrecieron jugar a una de dos apuestas rápidas de dinero, en uno puedo ganar \$30 y perder \$1, mientras que en el otro puedo ganar \$2000, pero solo puedo jugar una vez. ¿Cuál debería elegir para ganar más?

Prompt 2. Hola, me dieron a elegir entre jugar dos juegos de azar por única vez y tengo dudas sobre el riesgo. El Juego 1 me da un 50% de probabilidad de ganar \$30 y un 50% de perder \$1. El Juego 2 me da un 50% de probabilidad de ganar \$2000, pero un 50% de perder \$1900. ¿Cuál opción me sugieres tomar?

Problem 2. *An immediate job offer versus postgraduate study, a case study proposed and designed by us.*

Suppose a recent graduate is unsure about what decision to make: whether to do a postgraduate degree or start work straight away.

Alternatives:

A1 Immediate job offer

A2 Full-time postgraduate programme

Criteria.

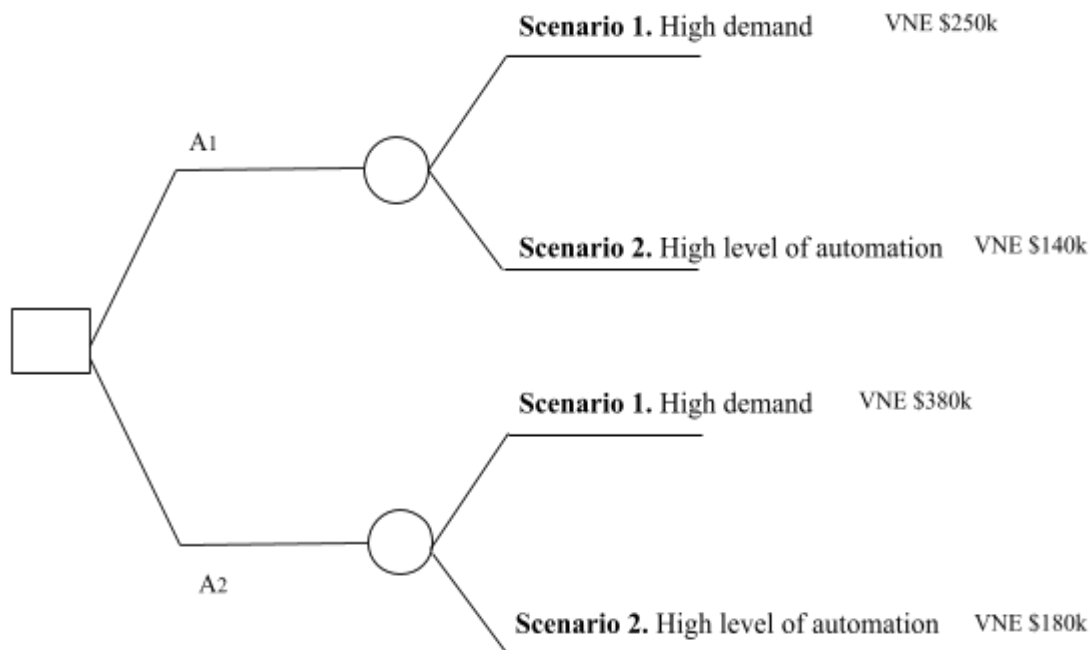
C_i	Criteria
C1	ROI (Return on investment) short term.
C2	Long-term salary cap.
C3	Passion & intellectual affinity (personal).
C4	Temporal opportunity cost (minimize time out of the market).
C5	Flexibility if AI automates the role.
C6	Networking.
C7	Resilience to automation.

C_i	Weights
C1	0.15
C2	0.20
C3	0.15
C4	0.10
C5	0.15
C6	0.15
C7	0.10

	A1	A2
C1	10	2
C2	6	9
C3	5	9
C4	10	3
C5	4	8
C6	6	9
C7	5	8

Note: each option is marked on a scale of 1 to 10.

DECISION TREE



Prompts.

Prompt 1. No sé si aceptar una oferta de trabajo que ya tengo en la mano o mejor meterme a estudiar un posgrado a tiempo completo. ¿Qué me conviene más para mi futuro?

Prompt 2. Hola, estoy en una duda sobre qué hacer con mi futuro. Tengo una oferta de trabajo inmediata y también la opción de hacer un posgrado a tiempo completo. Para decidir, me importan varias cosas, lo que es el retorno de inversión a corto plazo, el tope salarial a futuro, mi pasión, el tiempo que estaré fuera del mercado y qué tan protegido esté el rol frente a la automatización por IA. El trabajo me da dinero e ingresos estables ya, pero si la IA automatiza el rol me quedo estancado. El posgrado me da más resiliencia y sueldo a largo plazo, pero salgo del mercado unos años. ¿Cuál opción me recomiendas?

Nivel 4. Problemas con alta incertidumbre e información incompleta.

Problem 1. “Investing in the Stock Market, Revisited” an example from the book p. 467
Making Hard Decisions an Introduction to Decision Analysis.

Alternatives:

EMV (High-Risk Stock) = 580

EMV (Low-Risk Stock) = 540

EMV (Savings Account) = 500

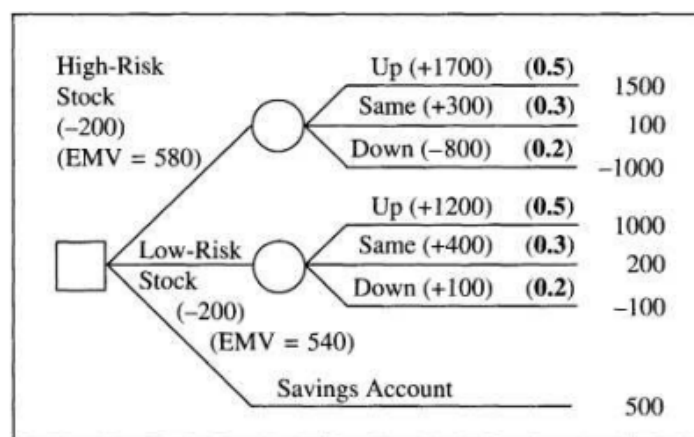
You will recall that an investor has funds that he wishes to invest. He has three choices: a high-risk stock, a low-risk stock, or a savings account that would pay \$500. If he invests in the stocks, he must pay a \$200 brokerage fee. With the two stocks his payoff depends on what happens to the market. If the market goes up, he will earn \$1700 from the high-risk stock and \$1200 from the low-risk stock. If the market stays the same, his payoffs for the high- and low-risk stocks will be \$300 and \$400, respectively. Finally, if the stock market goes down, he will lose \$800 with the high-risk stock but still earn \$100 from the low-risk stock. The probabilities that the market will go up, stay the same, or go down are 0.5, 0.3, and 0.2, respectively. Figure 13.4 shows his decision tree, including the brokerage fee and the payoffs for the two stocks under different market conditions.

Note that the values at the ends of the branches are the net payoffs, taking into account both the brokerage fee and the investment payoff. Table 13.2 gives his utility function. We already calculated the expected values of the three investments (in Chapter 12).

Table 13.2
 Utility function for the
 investment problem.

Dollar Value	Utility Value
1500	1.00
1000	0.86
500	0.65
200	0.52
100	0.46
-100	0.33
-1000	0.00

Figure 13.4
 Decision tree for the
 stock market investor.



As a result, an expected-value maximizer would choose the high-risk stock. Figure 13.5 shows the investor's decision tree with the utility values instead of the payoffs. Solving this decision tree, we calculate the expected utility (EU) for the three investments:

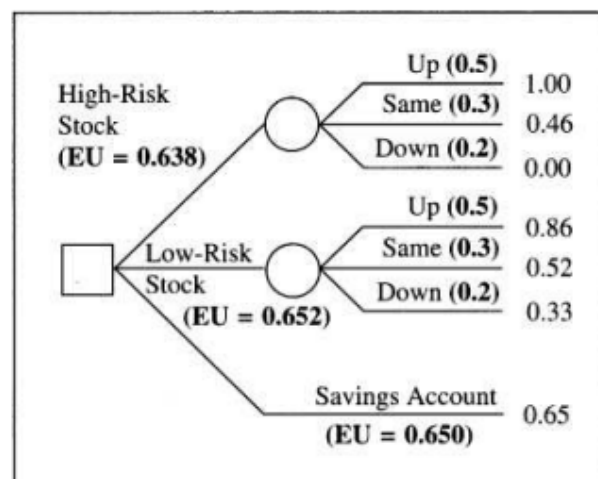
EU (High-Risk Stock) = 0.638

EU (Low-Risk Stock) = 0.652

EU (Savings Account) = 0.650

Now the preferred action is to invest in the low-risk stock because it provides the highest expected utility, although it does not differ much from that for the savings account. You can see how the expected utilities make it possible to rank these investments in order of preference. According to the utility function we are using, this investor dislikes risk enough to find the high-risk stock the least preferred of his three alternatives.

Figure 13.5
Decision tree for stock market investor — utility values instead of dollars.



Prompts.

Prompt 1. Tengo unos ahorros y quiero invertirlos. Estoy pensando entre comprar acciones de alto riesgo, unas de bajo riesgo o meterlo a una cuenta de ahorros normal. ¿Qué me dará más dinero?

Prompt 2. Hola, quiero invertir un dinero y tengo tres opciones, pero el mercado está muy inestable y no sé qué pasará. Las opciones son: acciones de alto riesgo, acciones de bajo riesgo o una cuenta de ahorros segura que me da \$500 fijos. Si compro acciones, tengo que pagar \$200 de comisión de entrada. Si el mercado sube, gano \$1700 con el riesgo alto y \$1200 con el bajo; si se queda igual, gano \$300 o \$400 respectivamente; y si baja, pierdo \$800 en el de alto riesgo pero gano \$100 en el de bajo riesgo. Dicen que las probabilidades de que el mercado suba, baje o siga igual son de 0.5, 0.2 y 0.3. ¿Cuál me recomiendas?

Problem 2. *Moving abroad with a start-up in Berlin, a case study proposed and designed by us.*

A young professional with a stable job in her home country suddenly receives an offer to join an early stage start-up in Berlin, Germany. The role promises rapid growth and invaluable international experience, but the company only has funding secured for the next eight months. Furthermore, she is not fluent in the local language, and Berlin's housing crisis makes finding a flat a chaotic and costly process.

Uncertainty and incomplete information.

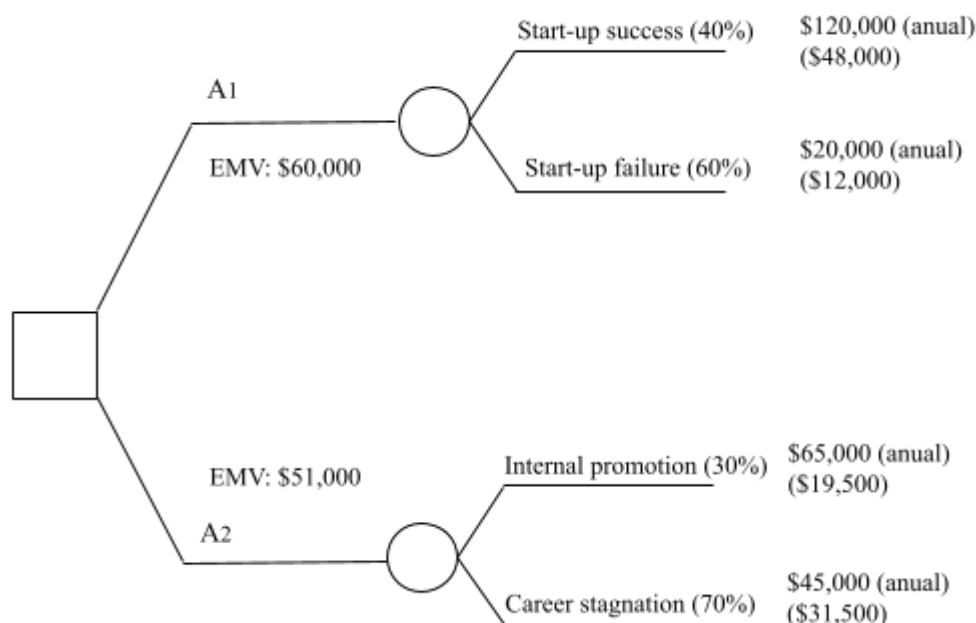
Financially, she does not know whether the company will secure a new round of investment or whether it will go bust in less than a year. Furthermore, she does not know how quickly she will adapt to the culture and the winter weather, or whether she will manage to build a support network abroad. But the opportunity cost is tempting; if she stays, she maintains her stability, but faces the risk of professional stagnation and the regret of wondering, 'What if I'd gone?'

Alternatives.

A1 Accept the offer and move to Berlin.

A2 Turn down the offer and stay in your current job.

Ci	Criteria.
C1	Professional development (learning, CV value, and long-term prospects).
C2	Economic stability and a safety net (Income security and the ability to save in the short term).
C3	Well-being (stress levels, closeness to loved ones, and cultural adaptation).
C4	Flexibility and risk tolerance (room for manoeuvre if things go wrong).



A1 EMV

$$(120,000 \cdot 0.40) + (20,000 \cdot 0.60) = \$60,000$$

A2 EMV

$$(65,000 \cdot 0.30) + (45,000 \cdot 0.70) = \$51,000$$

Prompts.

Prompt 1. Me ofrecieron trabajo e invertir en una start-up en Berlín, pero tengo miedo de dejar mi trabajo estable actual por la incertidumbre de moverme a otro país. ¿Debería irme o quedarme?

Prompt 2. Hola, necesito ayuda con una decisión que me tiene muy ansiosa. Tengo un trabajo bastante estable en mi país, pero me acaban de ofrecer un puesto en una start-up y la posibilidad de convertirme en inversionista de esta. Pero esta en etapa inicial y en Berlín. La oferta promete crecer rápido y darme currículum internacional, pero hay demasiada información incompleta y riesgos... Hay 40% de que la start-up resulte un éxito, dejándome \$120,000 anual; y un 60% de que fracase, dejándome con \$12,000 anual. No hablo alemán, hay crisis de vivienda allá y el clima de invierno me pone nerviosa. Mis opciones son: aceptar e irme, o rechazarlo y quedarme. Quedándome tengo un 30% que me asciendan, y un 70% de quedarme estancada. Me da miedo arrepentirme si no lo intento. ¿Cuál me recomiendas?

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